

NATIONAL TELECOMMUNICATION INSTITUTE (NTI)

Machine Learning Summer Training 2026

Implementation & Analysis of the **Adult Income Dataset**

PREPARED FOR

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ADULT INCOME DATA SET

Dataset Overview & Prediction Goal

What is the Data?

A demographic and employment dataset used to study factors associated with a person's income level, commonly applied in classification tasks.

The Target Variable

Predict the binary income class of individuals:

`<= 50K` or **`> 50K`**

What does it Contain?

The dataset tracks **12 key variables** describing individuals:

- Age
- Work class
- Education
- Marital status
- Occupation
- Relationship
- Race
- Sex
- Capital gain
- Capital loss
- Hours worked

Goal of the Dataset

Build classification models to predict whether an individual's annual income exceeds \$50K based on demographic & employment features.

Common benchmark for training predictive classifiers and evaluating model performance.

ADULT INCOME DATA SET

Dataset Characteristics & Target Distribution

Dataset Dimensions

32,561

Initial Rows / Instances

15 Columns (Metrics)

Target Variable: Income Column

Mapped as a **binary classification** problem to predict whether an individual's annual income exceeds \$50K:

Class 0: $\leq 50K$

Class 1: $> 50K$

Significant Class Imbalance

The dataset exhibits a strong skew, with the majority of individuals falling into the lower income bracket.

$\leq 50K$ (75.2%)

$> 50K$ (24.8%)

■ **Majority Class (75.2%):** ~24,486 individuals earning $\leq 50K$. Represents the baseline performance threshold for classification models.

■ **Minority Class (24.8%):** ~8,075 individuals earning $> 50K$. Requires robust evaluation metrics (F1-score, AUC-ROC) over simple accuracy.

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Data Cleaning & Preprocessing

Deduplication

3,465

Duplicate Records Removed

Redundant instances identified and permanently purged to ensure absolute data integrity and prevent model bias.

Feature Pruning

2

Redundant Columns Dropped

fnlwgt: Dropped due to lack of predictive value.

education: Dropped because education level is already accurately encoded by the **education.num** numerical feature.

Missing Values

2,192 Rows

7.53% of total dataset

Imputation Strategy

Retained rows with missing values in **workclass**, **occupation**, and **native.country** due to their unique, non-random income distributions.

- Replaced blank entries with explicit 'Unknown' or 'Other' category tags.
- Preserves valuable data footprint while preventing predictive bias in classifiers.

Exploratory Data Analysis (Numerical)

Age Profile

39.25 Years Avg

The average age is **39.25 years**, with the majority of demographic instances falling within the **28 to 48** age range.

Work Hours

40.6 Hrs/Week Avg

While the mean sits at **40.6 hours**, this operational feature exhibits a high outlier rate of **29.13%**.

Financial Features

\$0 Median Value

Both **capital.gain** and **capital.loss** are extremely skewed, with median values sitting at **\$0** for both.

Key Correlations with Income

The numerical metrics demonstrating the strongest statistical correlations to the binary target variable.

education.num
0.33

capital.gain
0.23

hours.per.week
0.22

age
0.21

Continuous features are highly predictive, with **education level (num)** standing as the single strongest demographic predictor of whether an individual earns above \$50K.

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Feature Engineering & Preprocessing

Categorical Grouping

Consolidated redundant categories to reduce noise in high-cardinality features. For instance, merged **'Husband'** and **'Wife'** into a single, unified **'Spouse'** relationship profile.

Government & Employment Classes

Simplified employment groups: combined **'State-gov'** and **'Local-gov'** into **'State/Local-gov'**, and consolidated marginal activities **'Without-pay'** and **'Never-worked'** into a unified **'No-work'** category.

Processing Pipeline

Constructed an automated sklearn column transformer to feed clean inputs directly into classification engines:

- Applies **median imputation** to preserve numeric footprints.
- Executes **standard scaling** for distance-based predictors.

One-Hot Encoding & Imputation

Categorical pipelines perform **most-frequent imputation** before expanding into binary vectors.

The transformed training matrix scales up to a highly informative dimensionality of **52 Features**.

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Modeling Strategy

Data Splitting

80% / 20% Train / Test

Split the processed dataset using **stratified sampling** to ensure both training and testing subsets precisely preserve the baseline target class distribution.

Algorithm Selection

7 Classification Models

Evaluated performance across diverse architectures: **Logistic Regression, KNN, Decision Tree, Naive Bayes, SVM, Random Forest, and Gradient Boosting.**

Optimization Pipeline

Utilized robust hyperparameter tuning techniques to maximize predictive performance:

- **5-Fold Cross-Validation** to guard against overfitting.
- **GridSearchCV** to exhaustively tune hyperparameters like max_depth, n_estimators, and learning_rate.

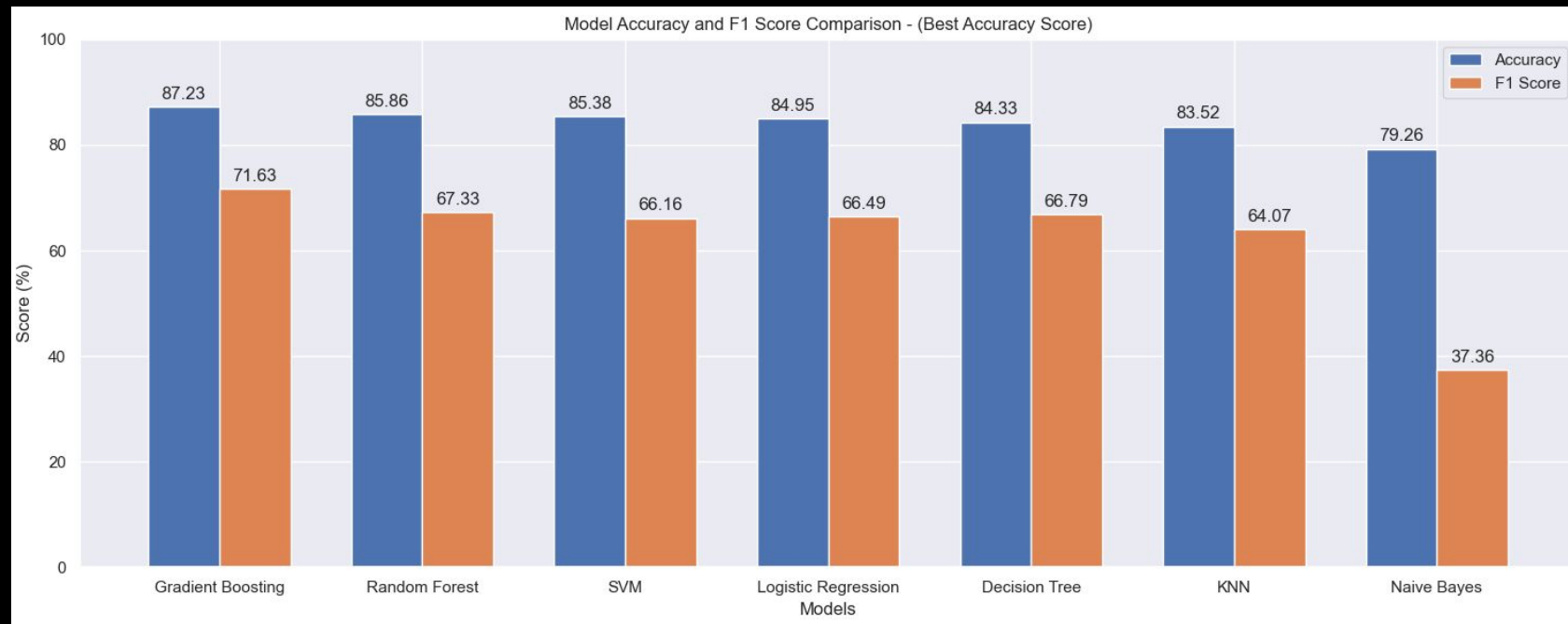
Evaluation Metric Priority

F1 Score Optimized

Prioritized the **F1 Score** as the primary tuning metric during cross-validation. This balances precision and recall, directly accounting for the heavy **75.2% to 24.8% class imbalance** in the dataset.

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Model Performance Comparison



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Model Performance Comparison



Thank You

Adult Income Data Set Analysis • Modeling Strategy & Performance Evaluation